

MSCI152 Coursework 2022

A bike-share rental company in Seoul is looking to analyse demand for rental bikes. They have brought you in, as a Business Analytics expert, to analyse their bike rental data to:

- Produce a simple dashboard in Excel that summarises the bike rental performance over the period the data covers;
- Construct a multiple linear regression model to predict demand for bike rentals and present the results of your analysis in a Management Report.

You have been provided with data in an Excel file “bikeRentalData.xlsx”. (Source: <http://data.seoul.go.kr/>) It contains 366 days of data from December 2017 to November 2018. The data description is as follows:

Total Rentals – Total number of bike rentals

Avg. Temp – Average temperature (in degrees Celcius °C)

Avg. Humidity – Average humidity (%)

Avg. Wind – Average windspeed (metres per second, m/s)

Avg. Visibility – Average visibility (in metres, m). Maximum visibility recorded is 2000m.

Avg. Dew Point Temp – Average dew point temperature each day (in degrees Celcius °C)

Avg. Solar Radiation – Average solar radiation (in mega joules per metre squared, MJ/m²)

Total Rain – Total rainfall (in millimetres, mm)

Total Snow – Total snowfall (in centimetres, cm)

Date – Date the observations were recorded

Season – Season the observations were recorded (Winter, Spring, Summer, Autumn)

Holiday – Records if day was a holiday or not (Holiday, No Holiday)

Operating Day – Records if the rental bike service was operating that day or not (Yes, No)

Requirements for the analysis

Conduct the following requested analyses:

1. **Bike rental dashboard.** Create a simple prototype dashboard in Excel that summarises the performance of the bike rental service over the period the data covers. It should include **a time-series plot** of the bike rentals. It should also include the average daily number of bike rentals in each season. You can include other charts, tables and key performance indicators (KPIs) in your dashboard that you think would interest the company.

The company expects that this dashboard will be easy to read, accessible and self-explanatory. However, they require a brief summary of the presented analysis including any insights that you can provide. This should be included in the management report as an appendix of up to a maximum of 2 pages.

2. Bike rental modelling. As for modelling, the company would like you to explain **daily bike rentals** via **multiple regression models** in the following way:

- Construct a multiple linear regression model that includes all quantitative data as independent variables (Full Model). This should be supported an exploratory data analysis appropriate for regression modelling. You should also determine whether the model violates any assumptions for regression modelling.
- Expand the Full Model above to include qualitative information on the season. Create dummy variables Spring, Summer and Autumn for each day that take the value 1 if the season is Spring, Summer and Autumn respectively. The dummy variables take the value 0 otherwise. You should determine whether the model violates any assumptions for regression modelling. Compare this model to the Full Model and determine which model is preferred.
- Build a third, improved, model utilising all relevant data. For this model you should:
 - Decide upon, and explain which variables should be included in the model and why.
 - Compare this model to your preferred choice from the earlier models.
 - Give a comprehensive interpretation of the improved model

This analysis should be included in the management report, up to a maximum of 8 pages in total. This page count includes the executive summary.

Your task

Use the information above and the data in the Excel spreadsheet to produce a dashboard and write a management report to the company on the requested analyses. Use any methods from the MSCI152 module that you think are useful and appropriate. However, you will not get any credit and may lose marks if you use methods that are not suitable.

You can assume that the company is familiar with all the statistics that have been covered in the course and so you do **not** need to explain any statistical terminology that you use, they simply require an expert to perform the analysis.

You are required to submit 2 elements: i) A prototype dashboard in Excel; and ii) A management report for your analysis that clearly explains the analyses you have done. Marks will be awarded for the choice and look of visualisations, quality (technical correctness) and clarity (quality of explanations) of your responses to the analysis requirements above. The weighting of the assessment marks are as follows:

- Dashboard and with supporting summary appendix (20% of the marks)
- Regression analysis of bike rentals (60% of the marks)
- Management report (20% of the marks)

You are expected to perform your analysis on the data provided using Excel. Using charts or analyses from other sources in place of the analyses you should be doing for the report will result in a mark of zero for that element of the report.

Report organisation and layout

The **maximum length** of the report is 10 pages (not counting the title page). You can use any readable font (i.e., not too small) and use single spacing. There is no penalty for a shorter report, although it is unlikely that you would be able to address all the issues properly in a very short report. You need to provide the information requested and explain the analysis you have done.

The report should follow the guidelines set out in the Lecture on Report Writing. However, since this is a short report:

- You do **not** need to start different sections on separate pages
- A contents page is **not** required. If you include a contents page it will count towards the total page limit.
- An appendix should be provided containing a brief summary of the prototype Excel dashboard.

Plagiarism

This coursework is to be done as an individual piece of work. All your analysis and the whole of your report must be your own work alone. Collusion between students and plagiarism are regarded as very serious offences and will be penalised severely.

Please note that software to detect plagiarism and collusion will be used.

Submission instructions

- Your report should be submitted to the **MSCI152 Moodle site** as two documents. An Excel spreadsheet containing the simple prototype dashboard and a Word or PDF document containing the management report. If possible, please submit your report as a **PDF file** (you should be able to save it in PDF format from Word). Alternatively, submit it as a **Word file** if that is not possible.
- The place to submit your file is in the **MSCI152 Moodle site: click on “Report” in the first section.**
- **Do not submit any other files.** Your report must be self-contained but can refer the Dashboard only.
- **The deadline for submission is 10am on Tuesday 24th January 2023**
- Remember to give yourself the opportunity to complete the task with time to spare. Leaving it to the last minute often results in a poor submission and a low mark. Late work without an agreed extension will be penalised according to the University rules (deduction of marks as shown in the assessment rules table for work up to 3 days late; a mark of 0 for work more than 3 days late). Extensions may, in certain **rare** circumstances, be given but to obtain an extension you must download the extension request form from the Management Science FAQ Moodle site. **You are very strongly advised to make back-up copies of your work during the project**, e.g. on your hard drive, a USB and/or on a cloud storage service – **loss of work due to computer problems will not be accepted as a reason for an extension.**